

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[2\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right] \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2. $\frac{-2 - 2i\sqrt{3}}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

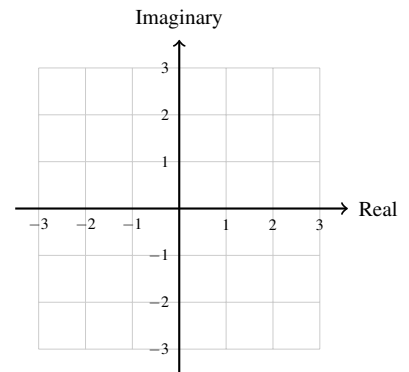
3. $\left[2 \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^8$

Solve. Write answers in rectangular form. (3 points)

4. $x^4 + 81 = 0$

Find the cube roots of the number. Leave answers in trig form. Graph each cube root as a vector in the complex plane. (3 points)

5. $2 - 2i\sqrt{3}$



ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Every line shown. Point splits are an estimate of partial credit (halves are possible).

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[2\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right] \left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]$

1. Multiply the r 's, add the θ 's. The sum is $\frac{11\pi}{12} = 165^\circ$, which is *not* on the unit circle, so use the sum formula.

$$\begin{aligned} r &= (2\sqrt{2})(\sqrt{2}) = 4 & \theta &= \frac{2\pi}{3} + \frac{\pi}{4} = \frac{8\pi}{12} + \frac{3\pi}{12} = \frac{11\pi}{12} \\ \cos \frac{11\pi}{12} &= \cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4} = \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{-\sqrt{2}-\sqrt{6}}{4} \\ \sin \frac{11\pi}{12} &= \sin \frac{2\pi}{3} \cos \frac{\pi}{4} + \cos \frac{2\pi}{3} \sin \frac{\pi}{4} = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} = \frac{\sqrt{6}-\sqrt{2}}{4} \\ 4 \left(\frac{-\sqrt{2}-\sqrt{6}}{4} + \frac{\sqrt{6}-\sqrt{2}}{4} i \right) &= \boxed{-\sqrt{2}-\sqrt{6} + (\sqrt{6}-\sqrt{2})i} \end{aligned}$$

Trap: $\cos(A+B) = \cos A \cos B - \sin A \sin B$ (minus), $\sin(A+B) = \sin A \cos B + \cos A \sin B$ (plus). QII check: $\cos$ negative, $\sin$ positive.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-2-2i\sqrt{3}}{\sqrt{2}-i\sqrt{6}}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

$$\begin{aligned} \text{Top: } r &= \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4+12} = 4 & \tan \theta &= \frac{-2\sqrt{3}}{-2} = \sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIII} \Rightarrow 240^\circ \\ \text{Bottom: } r &= \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{2+6} = \sqrt{8} = 2\sqrt{2} & \tan \theta &= \frac{-\sqrt{6}}{\sqrt{2}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ \end{aligned}$$

$$\begin{aligned} \frac{4 \text{cis } 240^\circ}{2\sqrt{2} \text{cis } 300^\circ} &= \frac{2}{\sqrt{2}} \text{cis}(240^\circ - 300^\circ) = \sqrt{2} \text{cis}(-60^\circ) = \sqrt{2} \text{cis } 300^\circ \\ &= \sqrt{2} \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = \boxed{\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}i} \end{aligned}$$

Trap: $\frac{2}{\sqrt{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$. $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$. Both parts of the top are negative: QIII, 240° (not 60°).

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[2 \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) \right]^8$

3. De Moivre: raise r to the 8th, multiply the angle by 8, then reduce by multiples of 2π .

$$\begin{aligned} 2^8 &= 256 & 8 \cdot \frac{7\pi}{12} &= \frac{56\pi}{12} = \frac{14\pi}{3} \\ \frac{14\pi}{3} - 4\pi &= \frac{14\pi}{3} - \frac{12\pi}{3} = \frac{2\pi}{3} \\ 256 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) &= 256 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = \boxed{-128 + 128\sqrt{3}i} \end{aligned}$$

Trap: $\frac{7\pi}{12}$ is not a unit-circle angle, but $8 \cdot \frac{7\pi}{12}$ reduces to one. Don't reach for the sum formula before multiplying.

Solve. Write answers in rectangular form. (3 points)

Problem: $x^4 + 81 = 0$

4. Isolate x^4 , write the constant in trig form, take fourth roots, convert each.

$$x^4 = -81 = -81 + 0i \Rightarrow r = 81, \theta = 180^\circ \quad x = (81 \text{ cis } 180^\circ)^{1/4}$$

$$\sqrt[4]{81} = 3 \quad \frac{180^\circ}{4} = 45^\circ \quad \text{spacing } 90^\circ \quad x = 3 \text{ cis } 45^\circ, 3 \text{ cis } 135^\circ, 3 \text{ cis } 225^\circ, 3 \text{ cis } 315^\circ$$

$$3 \text{ cis } 45^\circ = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i \quad 3 \text{ cis } 135^\circ = -\frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i$$

$$3 \text{ cis } 225^\circ = -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i \quad 3 \text{ cis } 315^\circ = \frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i$$

$$x = \pm \frac{3\sqrt{2}}{2} \pm \frac{3\sqrt{2}}{2}i \quad (\text{all four sign combinations})$$

Trap: $x = \pm 3i$ is wrong: $(3i)^4 = 81i^4 = 81$, not -81 . An even power of a pure imaginary is always positive real, so the roots sit between the axes ($45^\circ, 135^\circ, \dots$).

Find the cube roots of the number. Leave answers in trig form. Graph each cube root as a vector in the complex plane. (3 points)

Problem: $2 - 2i\sqrt{3}$

5. Convert, then cube roots. $\sqrt[3]{4}$ does not simplify, so leave it.

$$r = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

$$\tan \theta = \frac{-2\sqrt{3}}{2} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$$

$$\sqrt[3]{4} \quad \text{first: } \frac{300^\circ}{3} = 100^\circ \quad \text{spacing } 120^\circ$$

$$\sqrt[3]{4} \text{ cis } 100^\circ, \sqrt[3]{4} \text{ cis } 220^\circ, \sqrt[3]{4} \text{ cis } 340^\circ$$

The graph: three equal vectors of length $\sqrt[3]{4}$ (between 1 and 2), 120° apart (the "peace sign").

Trap: divide the angle, not the radius: $\sqrt[3]{4}$, not $\frac{4}{3}$. All angles stay positive and under 360° .

